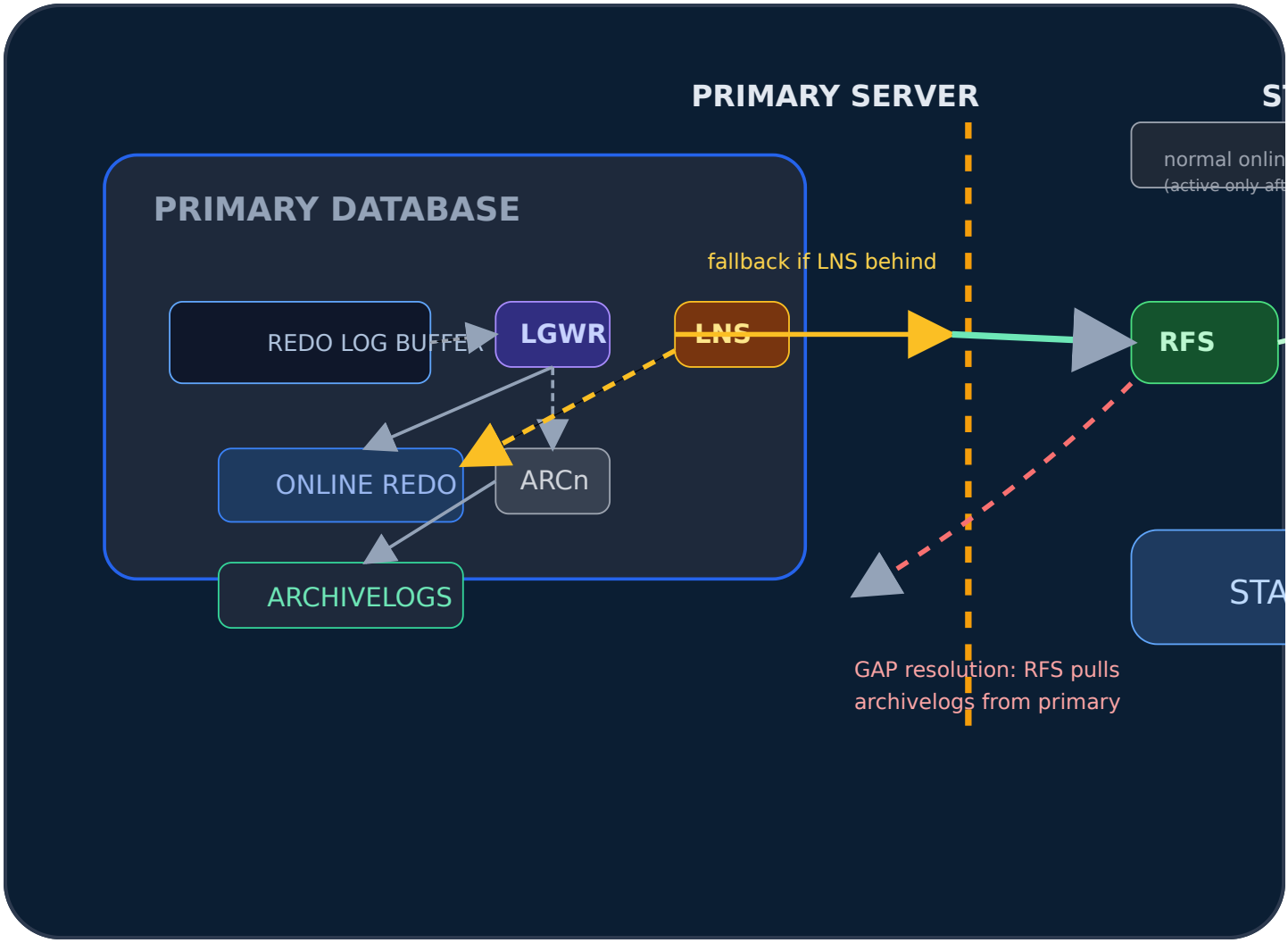
Oracle Data Guard · Physical Standby Deep Dive

Physical standby architecture – component flow

LNS (Log Network Server) · **RFS** (Remote File/Fetch Service) · **MRP** (Managed Recovery Process) · **SRL** (Standby Redo Logs)



Primary side processes

Standby side processes

- **LGWR** – writes redo to online redo logs (cyclic).
- **LNS** (log network server) – reads redo buffer (or falls back to online redo if behind) and ships over Oracle Net.
- **ARCn** – archiver, creates archivelogs, used by RFS for gap resolution.

- **RFS** (remote file/fetch service) – receives redo from LNS, writes to **standby redo logs (SRL)**. Can fetch missing archivelogs (gap resolution).
- **MRP** (managed recovery process) – applies redo from SRL to standby database.

⚡ **LNS fallback:** If LNS cannot keep up with LGWR, it reads directly from online redo logs (dashed gold arrow).
⚡ **Gap resolution:** When standby detects missing logs, RFS contacts primary ARCn and pulls archivelogs automatically (red dotted line) — but if RMAN deleted them, manual intervention is needed.

Standby redo logs (SRL) vs. online redo logs

Online redo logs (primary & standby)

- Cyclic, written by LGWR.
- On standby: remain idle until role switch (standby becomes primary).
- Not used for media recovery while in standby role.

Standby redo logs (SRL)

- Special logs created `ALTER DATABASE ADD STANDBY LOGFILE`.
- RFS writes incoming redo directly to SRL.
- MRP reads SRL to apply changes.
- **Must exist on both primary and standby for seamless role transition (switchover/failover).**

⚠ **Interview tip:** Many DBAs forget SRL on primary. After failover, the new primary (old standby) expects SRL on the demoted primary. Without SRL, you cannot perform a clean switchover.

⚠ RMAN-08137: WARNING – archived log not deleted

```
RMAN-08137: WARNING: archived log not deleted,
needed for standby or upstream capture process
```

RMAN refuses to delete archivelogs because they are still required by a Data Guard standby, GoldenGate extract, or an obsolete capture process. If ignored, archive log destinations fill up and databases may hang.

□ Why does this happen (even without Data Guard/GoldenGate)?

- **Orphaned capture processes** remain registered in `DBA_CAPTURE` from old replication.
- **Deletion policy** set to `APPLIED ON STANDBY` or `SHIPPED TO STANDBY` but standby is down or lagging.
- **Archive gap** – standby missing logs; they are preserved automatically.

□ METHOD 1: Check archive dependencies & policy

```
-- 1. Detect any archive gap between primary and standby
SELECT THREAD#, LOW_SEQUENCE#, HIGH_SEQUENCE# FROM V$ARCHIVE_GAP;

-- 2. Which archived logs are still not deleted?
SELECT SEQUENCE#, FIRST_TIME, DELETED FROM V$ARCHIVED_LOG WHERE
DELETED='NO';

-- 3. Show current RMAN archivelog deletion policy
RMAN> SHOW ALL;
```

If policy is too strict (e.g., `APPLIED ON STANDBY`), temporarily relax it:

```
RMAN> CONFIGURE ARCHIVELOG DELETION POLICY TO SHIPPED TO STANDBY;
RMAN> DELETE NOPROMPT ARCHIVELOG ALL COMPLETED BEFORE 'SYSDATE-1';
-- restore later
RMAN> CONFIGURE ARCHIVELOG DELETION POLICY TO NONE;
```

□ METHOD 2: Unused capture processes & standby lag

```
-- List any capture processes (GoldenGate/Streams)
SELECT CAPTURE_NAME, STATUS FROM DBA_CAPTURE;

-- If unused, unregister (GoldenGate example)
GGSCI> DBLOGIN USERIDALIAS <alias>
GGSCI> UNREGISTER EXTRACT <extract_name> DATABASE

-- Check Data Guard apply status
SELECT PROCESS, STATUS, SEQUENCE# FROM V$MANAGED_STANDBY WHERE PROCESS IN
('MRP0','RFS');

-- Verify logs applied on standby destinations
```

```
SELECT DEST_ID, APPLIED_SCN FROM V$ARCHIVE_DEST_STATUS WHERE
STATUS='VALID';
```

If standby is lagging, wait until MRP catches up or troubleshoot network/recovery.

METHOD 3: Proactive prevention

- **Monitor free space** in fast recovery area:
  `SELECT name, space_limit - space_used AS free_mb FROM V$RECOVERY_FILE_DEST;`
- **Set up alerts** when apply lag exceeds threshold (e.g., using Enterprise Manager or custom scripts).
- **Audit registered capture processes** monthly – remove obsolete entries.
- **In 11g/12c and later**, use `CONFIGURE ARCHIVELOG DELETION POLICY TO APPLIED ON ALL STANDBY` after ensuring all standbys are healthy.

> **Best practice:** Integrate archive deletion policy with backup retention; ensure it aligns with standby requirements. Use `BACKUP ... DELETE INPUT` carefully.

Gap resolution – when automation fails

RFS automatically fetches missing archivelogs from primary *if they still exist on disk.* But if RMAN or a backup job deleted the necessary archivelogs, you must intervene:

1. Identify gap sequence:
   `SELECT * FROM V$ARCHIVE_GAP;`
2. Restore missing logs from backup to primary (or copy from another source).
3. Register them via `ALTER DATABASE REGISTER LOGFILE '/path/...'`.
4. RFS will detect and ship them automatically.

! **Common cause of RMAN-08137:** Archive logs are deleted on primary before being shipped/applied on standby. Tune backup scripts with: `BACKUP ARCHIVELOG ALL NOT BACKED UP 1 TIMES DELETE INPUT;` and ensure deletion policy includes `APPLIED ON STANDBY`.

Interview / real-world Q&A

- **Q: Does standby need its own redo logs? Why SRL?**
  A: Yes, SRL separate from online redo. If standby is open (active Data Guard) it generates its own redo. SRL keeps incoming changes isolated.
- **Q: What if I don't configure SRL on primary?**
  A: Switchover will fail; primary cannot become a standby without SRL.
- **Q: Why does RMAN-08137 appear even if I have no standby?**
  A: Check for orphaned capture processes in DBA_CAPTURE, or a previously registered Data Guard broker configuration.
- **Q: Can I force delete logs causing RMAN-08137?**
  A: Only after confirming no replication/standby requires them, or after disabling the capture/unregistering. Use `DELETE ARCHIVELOG ... FORCE` as last resort, but risk breaking DR.

⚡ Based on real Data Guard training · LNS · RFS · MRP · gap resolution · RMAN-08137 explained